

How Far Does a Litter Detector Travel?

Cross-Domain Urban Waste Detection Across Handheld, Dashcam and Drone Imagery

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Abstract—Automated litter monitoring is usually evaluated within a single imaging domain, yet real deployments mix handheld reports, vehicle cameras and drones. We study how much urban waste detectors degrade across three public domains TACO (handheld), RoLID-11K (dashcam) and UAVVaste (drone) under a single audited evaluation protocol. Our curation uncovered seven defects in the released datasets, most notably video-level leakage in RoLID-11K’s official splits, where 58.2% of test images share a source video with training; a model trained with that overlap scores +21% AP_{50} on the same test set. We compare Faster R-CNN, an FCN segmenter and CAM-based interpretability against a YOLOv11 reference, and measure a cross-domain degradation of 82% mAP_{50} , dominated by object scale (median 22 px in dashcam imagery, below the smallest FPN anchor). Raising input resolution to 1024 px recovers up to 8.3 AP_{50} and 5.8 AP_S points. Code, weights and audited splits are released.

Index Terms—litter detection, cross-domain evaluation, dataset auditing, small objects

I. INTRODUCTION

Uncollected urban waste harms public health, tourism and municipal budgets, and its monitoring still relies on manual inspection with limited coverage. Computer vision promises continuous, low-cost monitoring, and recent work has produced public datasets for litter detection in complementary settings: street-level handheld photographs (TACO [1]), forward-facing dashcams (RoLID-11K [2]), drone imagery (UAVVaste [3]) and waste-truck fisheye cameras (StreetView-Waste [4]). Each benchmark, however, is evaluated *in-domain*; whether a detector trained on one viewpoint remains useful on another the situation any city deploying mixed sensors faces has not been measured.

This paper asks: *how much does urban litter detection degrade across imaging domains, and what matters most to close the gap architecture, input resolution, or training data hygiene?* Answering it required more than training detectors. Auditing the three datasets surfaced seven defects (Section III), including video-level leakage in RoLID-11K’s official splits that inflates published baselines, basename collisions and inconsistent EXIF annotation frames in TACO and UAVVaste. We therefore built an audited curation pipeline with frozen,

group-aware splits, and evaluate every model against the same ground truths.

Our contributions are: (i) a documented audit of three public litter datasets, with corrected and released splits; (ii) the first cross-domain evaluation matrix across handheld, dashcam and drone litter imagery; (iii) a quantification of the leakage inflation in RoLID-11K’s official protocol (+21% AP_{50} from scene memorization); (iv) a controlled comparison of classical fine-tuned detectors, segmentation and interpretability techniques against a modern real-time reference, including a small-object resolution ablation and a class-granularity (6 vs. 5 material classes) study.

II. RELATED WORK

Litter datasets. TACO [1] provides 1,500 handheld images with 60 litter categories and segmentation polygons. RoLID-11K [2] contributes 11.5k dashcam frames with a single *litter* class and pronounced small-object statistics. UAVVaste [3] covers low-altitude drone views. StreetView-Waste [4] targets waste-truck fisheye cameras; its download endpoints were unavailable during this work. Detect-waste [5] harmonized several of these sources; ZeroWaste [6] addresses industrial sorting, a domain we deliberately exclude.

Detection under constraints. TrashDet [7] searches TinyML detectors on TACO, reaching 19.5 mAP_{50} on a five-class subset evidence that absolute scores in this problem are structurally low, which motivates our focus on *relative* cross-domain quantities. RoLID’s authors benchmark modern detectors, with transformer heads (CO-DETR) leading YOLO variants by over 20 AP_{50} points, consistent with the extreme small-object regime.

Methods. We build on Faster R-CNN [8] and its RPN, fully convolutional segmentation [9], class-activation interpretability [10], [11], [12], YOLOv11 [13] as the real-time reference, TIDE [14] for error decomposition and iterative stratification [15] for group-aware splitting.

III. DATASETS, AUDIT AND CURATION

We ingest the three in-scope datasets through a medallion pipeline (raw→bronze→silver→gold) with per-layer manifests

TABLE I
THE THREE DOMAINS AFTER CURATION. MEDIAN OBJECT SIDE
MEASURED ON THE 1280-PX TRAINING POOL.

Dataset	Viewpoint	Images	Median obj.	% small
TACO	handheld	1,500	171 px	25
RoLID-11K	dashcam	11,564	22 px	84
UAVVaste	drone	772	72 px	66

and SHA-256 lineage; Figure 1 overviews the whole flow, from verified ingestion to the evaluation contract.

Table I summarizes the domains and Figure 2 shows the axis that separates them: object scale.

Audit findings. Systematic checks (perceptual hashing, EXIF-frame verification, split cross-referencing) surfaced seven defects we correct or document: (H1) TACO basenames collide across batches (92% of images share a filename with another batch); (H2) $\sim 37\%$ of TACO images carry pending EXIF rotations, with boxes annotated on the *rotated* frame; (H3) RoLID’s release folders do not encode capture hour; (H4) **RoLID’s official splits leak at the video level:** 22 of 84 source videos span multiple splits and 58.2% of test frames share a video with training near-duplicate frames (87% visual redundancy by perceptual hash) make this scene leakage, inflating published baselines; (H5) one image is duplicated between official val and test; (H6) UAVVaste’s official split also contains one near-duplicate pair across train/test; (H7) annotation frames are inconsistent across datasets TACO annotates on the EXIF-rotated frame while five UAVVaste handheld photos annotate the raw frame so pixels are materialized per-image in the frame that matches the declared dimensions, with EXIF metadata stripped.

Curation. Silver applies five corrections (duplicate removal from val, phantom-category removal, full-path identity, degenerate-box and MD5 flags), defines a two-level taxonomy *litter* (1 class, all datasets) and TACO material classes in two parallel variants, **A: 6 classes** (glass included, marginal at 38 projected test instances) and **B: 5 classes** (glass excluded) and freezes four split schemes with group-aware units: pHash visual groups for TACO/UAVVaste and *source video* for RoLID. TACO splits use iterative stratification [15]; every class lands within 0.1 pp of the 15% test target. For RoLID we keep *both* the official (leaky, for comparability) and a by-video (0% leakage) scheme, enabling a paired quantification of the inflation.

IV. METHODS

P0 - fine-tuned classical pipeline. (1) *Faster R-CNN* (ResNet-50-FPN, COCO-pretrained) under two transfer strategies frozen backbone vs. unfreezing `layer4` with linear warmup, gradient clipping and early stopping on validation AP_{50} . (2) *FCN-ResNet18* for binary litter/background segmentation on TACO’s rasterized polygons (1×1 conv + $\times 32$ transposed conv with bilinear init, class-weighted cross-entropy). (3) *VGG-16 transfer learning* on 224-px litter/background

crops (two strategies $\times$ two learning rates), providing the substrate for CAM [10], Grad-CAM [11] and Guided Backpropagation [12] interpretability panels. (4) An *anchor-coverage analysis*: FPN base anchors (32-512 px) against the measured object-size distributions, plus RPN objectness maps from the trained dashcam model a mechanistic account of the small-object gap.

P1 - modern reference. YOLOv11n [13] trained per domain at 640 px (and 1024 px for the resolution ablation), used to fill the 3×3 cross-domain matrix, the official-vs-by-video leakage pair, and the 6-vs-5-class variant comparison.

Evaluation contract. A single source of metric truth: predictions from every model are evaluated with COCO tooling against the same frozen ground truths; test is touched once per experiment with best-validation weights; saved predictions feed TIDE [14] error decomposition.

V. EXPERIMENTAL SETUP

Nineteen runs (11 P0, 8 P1) executed on two lanes: Apple M-series (MPS) for VGG/FCN/YOLO and CUDA (Colab A100) for Faster R-CNN and the heavier YOLO runs profiling showed healthy Faster R-CNN iterations cost ~ 95 s on MPS (variable input shapes force per-shape kernel recompilation), versus ~ 0.3 s on CUDA; an hour-budget guardian projected total cost after the first epoch and routed unviable runs automatically. Budgets: generous epoch caps (50-150) with early stopping (patience 8-30) and validation curves; seeds fixed; per-epoch checkpoints with resume. Images are materialized at max side 1280 (Lanczos, JPEG q95) in the annotation frame (H7). Figure 3 shows the validation trajectories of the eight YOLO runs: clean convergence, early stopping at work, and the separation by domain and resolution.

Multiclass exports exclude *poisoned negatives* images whose only objects belong to excluded classes rather than training them as background (1/38/60 images for the 1/6/5-class variants).

VI. RESULTS

Every number below is generated by the consolidated evaluation notebook from saved test predictions against the frozen ground truths; no metric is transcribed by hand.

A. Cross-domain matrix

Figure 4 presents the central result. In-domain, the three YOLOv11n models average 0.639 AP_{50} ; across domains this collapses to 0.118 an **82% degradation**. The matrix is strongly asymmetric: the dashcam domain is an island ($AP_{50} \leq 0.064$ in either direction), while handheld $\rightarrow$ drone transfers surprisingly well (0.527, retaining 68% of the drone in-domain score). This pattern indicts *object scale* rather than appearance alone: TACO and UAVVaste objects (median 171 and 72 px) both lie within the FPN anchor range, whereas RoLID’s 22-px median falls below the smallest 32-px anchor consistent with our anchor-coverage analysis and with AP_S collapsing hardest (diagonal AP_S : 0.056 / 0.279 / 0.374).

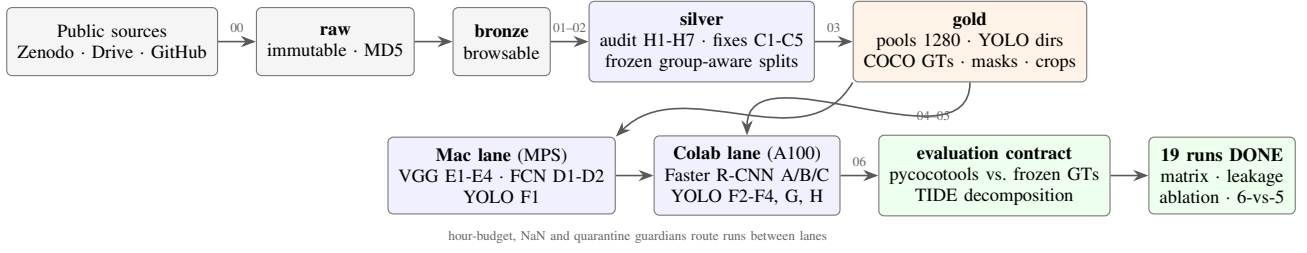


Fig. 1. End-to-end pipeline. Data flows through a medallion architecture with SHA-256 lineage (arrow labels: notebook numbers); training runs on two lanes governed by automatic guardians; every model is evaluated under a single metric contract.

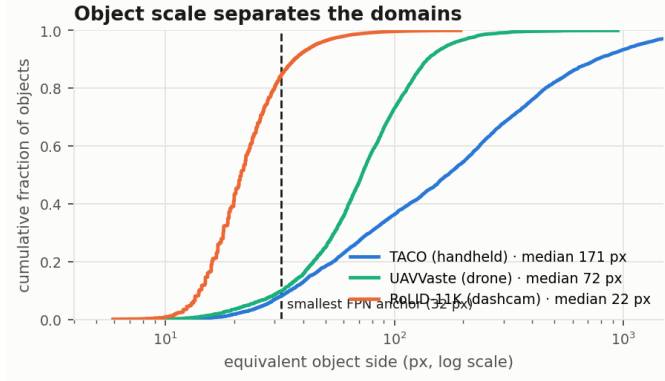


Fig. 2. Cumulative distribution of object side per domain. RoLiD’s median (22 px) falls below the smallest 32-px FPN anchor: the regime that explains the cross-domain matrix and motivates the resolution ablation.

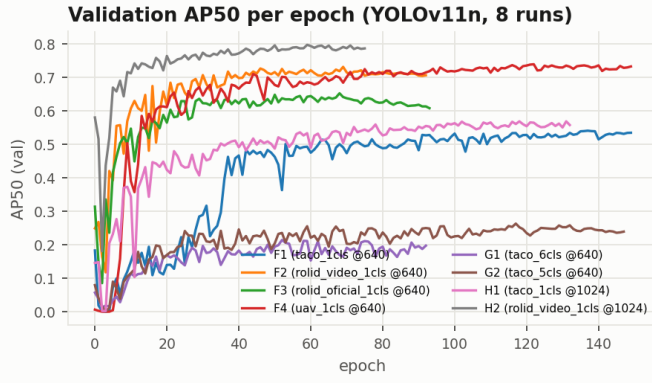


Fig. 3. Validation AP_{50} per epoch for the eight YOLOv11n runs.

B. Leakage inflation in RoLiD-11K

Table II pairs the official (58% video leakage) and by-video (0%) protocols. On the *same* clean by-video test set, the model trained on the official split which saw other frames of those very test videos scores 0.844 AP_{50} versus 0.695 for the cleanly trained model: a **+21% relative inflation** attributable to scene memorization, not generalization (87% of intra-video frame pairs are visually redundant by perceptual hash). Published baselines on the official protocol should be read with this bias in mind; group-aware splitting removes it at zero cost.

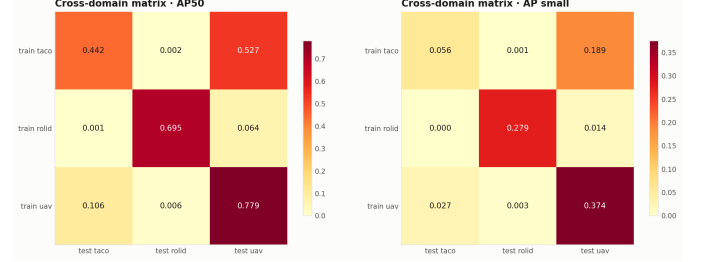


Fig. 4. Cross-domain matrix on test (YOLOv11n @640): AP_{50} (left) and AP_5 (right). Rows: training domain; columns: test domain. In-domain mean 0.639 vs. cross-domain 0.118.

TABLE II
LEAKAGE QUANTIFICATION ON RoLiD-11K (TEST AP_{50} , YOLOv11n @640).

Training split	Test set	AP_{50}
official (58% leak)	official	0.620
by-video (0% leak)	by-video	0.695
official (58% leak)	by-video	0.844

C. Families under the same protocol

Table III compares Faster R-CNN (41.3 M parameters) and YOLOv11n (2.6M) on identical splits and ground truths. On single-class tasks the 16 $\times$ smaller model stays within 1-4 AP_{50} points (0.695 vs. 0.707 on dashcam; 0.442 vs. 0.483 on handheld), supporting lightweight deployment; on material classification the gap widens to 5-9 points. Unfreezing layer4 lifts Faster R-CNN by +3.7 AP_{50} over the frozen backbone (0.483 vs. 0.446). Our 5-class YOLO (0.192 m AP_{50}) is consistent with TrashDet’s 19.5 on a comparable TACO subset [7], confirming that absolute scores in this problem are structurally low. The complementary P0 models complete the picture: FCN-ResNet18 reaches 0.577 test IoU on binary litter segmentation, and the VGG-16 crop classifier attains 97.8% test accuracy, providing the substrate for the CAM/Grad-CAM evidence that the network attends to the object rather than context (Figure 5).

D. Resolution and class granularity

Table IV shows the 640 $\rightarrow$ 1024 ablation. Gains concentrate exactly where objects are smallest: +8.3 AP_{50} and +5.8 AP_5 on dashcam, versus +3.2 and +3.9 on handheld resolution is

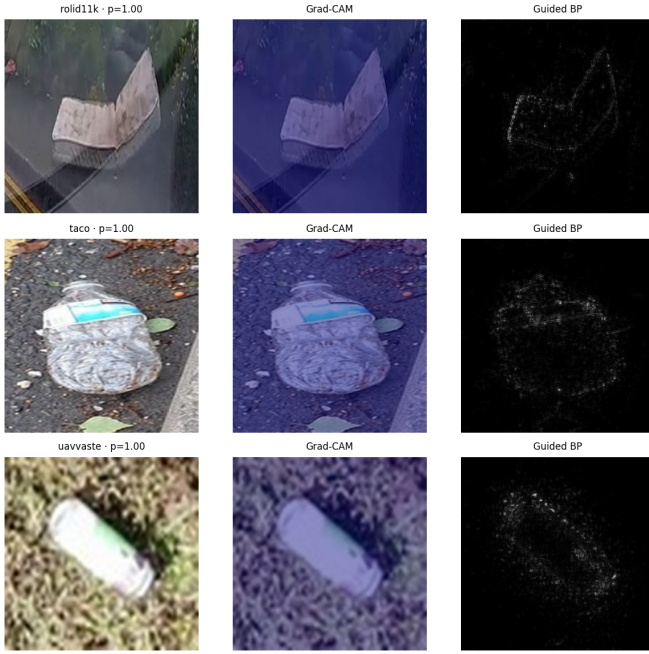


Fig. 5. Crop-classifier interpretability: one row per domain (dashcam, handheld, drone) with Grad-CAM and Guided Backpropagation. Activation concentrates on the object even in 22-px crops.

TABLE III
MODEL FAMILIES ON IDENTICAL SPLITS (TEST METRICS). FRCNN = FASTER R-CNN R50-FPN, LAYER4 UNFROZEN.

Data	Model	AP ₅₀	AP	AP _S
TACO 1-cls	FRCNN	0.483	0.323	0.073
	YOLOv11n	0.442	0.318	0.056
RoLID by-video	FRCNN	0.707	0.277	0.281
	YOLOv11n	0.695	0.279	0.279
TACO 6-cls	FRCNN	0.246	0.154	0.029
	YOLOv11n	0.157	-	-
TACO 5-cls	FRCNN	0.244	0.155	0.024
	YOLOv11n	0.192	-	-

the cheapest effective lever for the small-object regime. For class granularity, excluding the marginal glass class (38 test instances, AP₅₀ 0.049 when kept) improves the global 5-class score to 0.192 vs. 0.157 for 6 classes, with a mean +0.014 AP₅₀ on the five shared classes (largest on metal, +0.077): keeping a severely under-represented class taxes its siblings.

E. Error decomposition

Figure 6 decomposes the error with TIDE across all 13 detectors. Three regimes emerge. Multiclass models are dominated by *classification* error (up to 24.6 dAP in the 5-class YOLO, against only 1.7 for localization): finding litter is not the bottleneck naming its material is. Single-class handheld models split their error between background false positives and missed objects (~ 11 -13 dAP each). In-domain dashcam models, counter-intuitively, miss little (dAP_{Miss} ≤ 3.9): the

TABLE IV
RESOLUTION ABLATION (YOLOv11n, TEST).

Domain	AP ₅₀		AP _S	
	640	1024	640	1024
TACO	0.442	0.474	0.056	0.096
RoLID	0.695	0.779	0.279	0.337

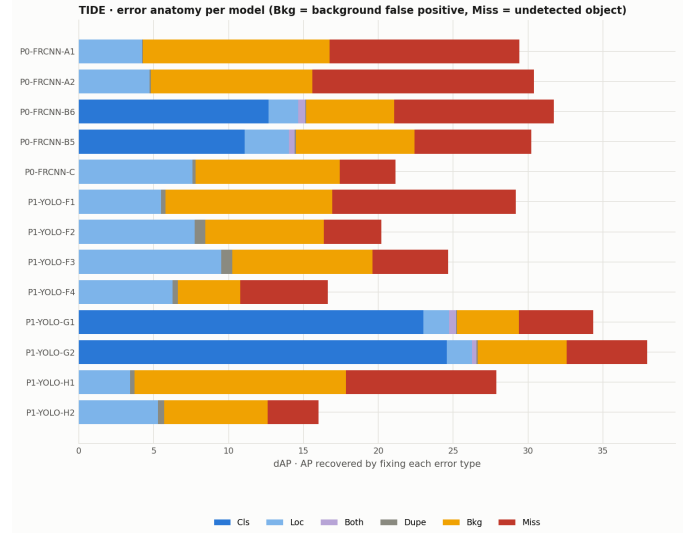


Fig. 6. TIDE error decomposition per model (dAP recovered by fixing each error type). Cls dominates multiclass models; Bkg+Miss dominate single-class ones.

22-px objects *are* found, and the residual error is dominated by background confusions evidence that the cross-domain collapse of Figure 4 stems from representations, not from an unrecoverable recall floor.

F. Qualitative out-of-distribution results

Figure 7 shows the batched demo (notebook 07) on a session of ten images downloaded from the internet, unrelated to the three datasets. Under conservative thresholds (0.25 for YOLO, 0.5 for Faster R-CNN), the quantitative patterns reappear out of distribution: Faster R-CNN proposes more candidates than YOLOv11n on all ten images (211 vs. 87 detections in total; 45 vs. 15 on the densest scene), the FCN estimates 3-16% litter pixels per scene, and the VGG classifier confirms every crop with P(litter) ≥ 0.99 , Grad-CAM attending to the object rather than its context. These images are illustrative evidence, not a metric: they carry no annotations.

VII. DISCUSSION AND LIMITATIONS

Three findings organize the evidence. **Litter detectors do not travel.** The 82% cross-domain drop, with dashcam isolated and handheld $\leftrightarrow$ drone partially compatible, is explained primarily by object scale: domains whose object sizes fall within the pretraining anchor range exchange some knowledge; the 22-px dashcam regime exchanges none. For municipal deployments mixing sensors, per-domain fine-tuning

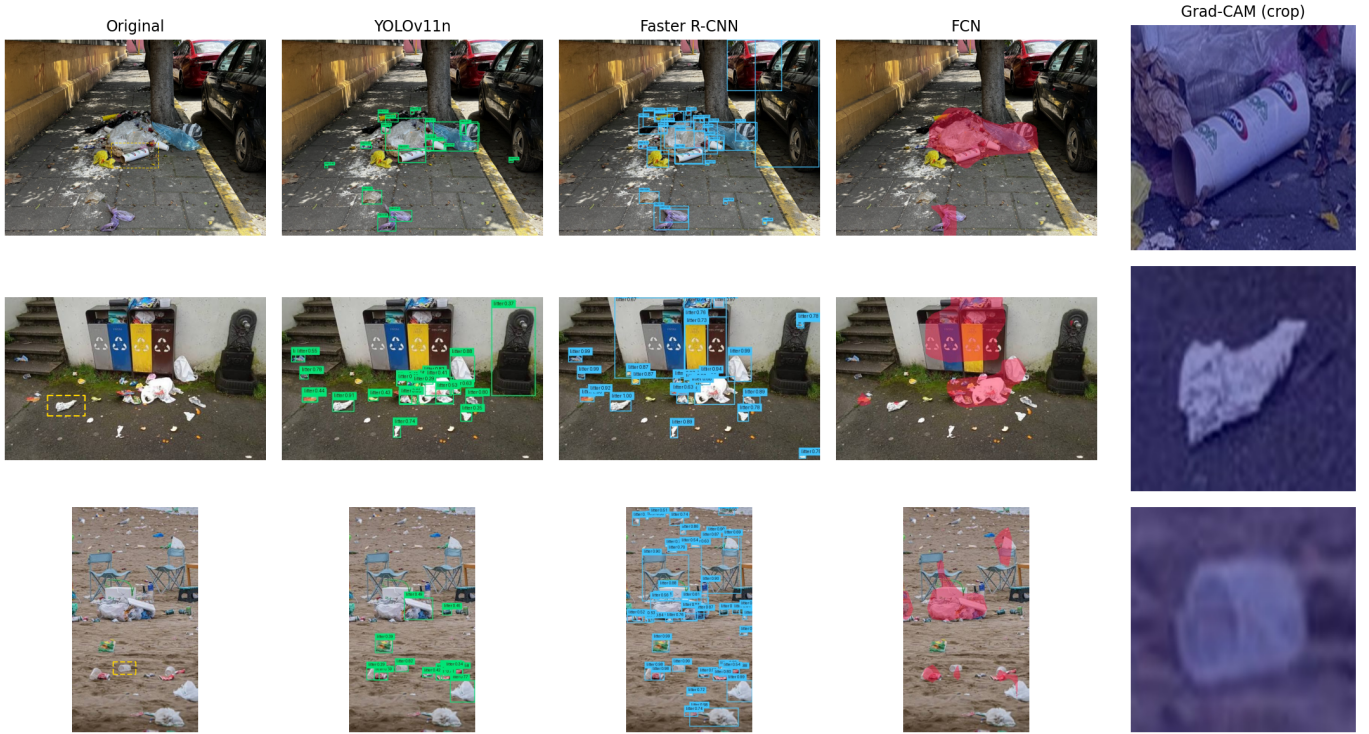


Fig. 7. Qualitative demo on out-of-distribution web images (three of the ten in the session): original with the Grad-CAM crop origin marked with a dashed box, YOLOv11n and Faster R-CNN detections, FCN litter mask and the crop classifier’s Grad-CAM explanation. Images randomly downloaded from the internet, with no embedded authorship metadata, used for illustration only; they will be replaced by our own captures (Lima) in a future revision.

or at minimum resolution adaptation remains mandatory; a single universal litter model is not yet realistic. **Split hygiene changes conclusions.** The official RoLID protocol rewards scene memorization (+21% AP_{50}); results published on it are optimistic by roughly that margin, and the fix (group-aware, by-video splitting) costs nothing. We argue dataset releases with video or burst structure should ship group identifiers. **The cheap levers work.** 1024-px input buys back +8.3 AP_{50} on dashcam; a 2.6M-parameter detector performs within a few points of a 41.3M one on single-class litter detection; and TIDE locates the multiclass bottleneck in material classification rather than localization suggesting a decoupled design (class-agnostic detector plus crop classifier, where our VGG substrate reaches 97.8% accuracy) over ever finer detection heads. Limitations we can already state: absolute mAP in this problem is structurally low [7]; our compute budget precluded transformer detectors (CO-DETR-class) which lead RoLID’s in-domain benchmark; StreetView-Waste could not be included (endpoints unavailable); our own OOD capture (Lima) remains future work, as does temporal aggregation over dashcam video natural given the video-level structure our audit exposed.

VIII. CONCLUSION

We audited three public litter datasets, released corrected group-aware splits, and measured under a single evaluation contract how far litter detectors travel across handheld, dash-

cam and drone imagery. The answers are concrete: an 82% mAP_{50} drop out of domain with dashcam fully isolated; a +21% AP_{50} inflation hiding in RoLID-11K’s official splits; up to +8.3 AP_{50} recoverable by input resolution alone; and a 16 $\times$ smaller modern detector within a few points of Faster R-CNN on single-class litter. Error decomposition shows the remaining battles are background discrimination and material classification, not localization. The full pipeline from check-summed ingestion to generated tables plus weights and a live demo are public; transformer detectors, temporal aggregation over dashcam video and our planned Lima out-of-distribution capture are the natural next steps.

REPRODUCIBILITY

Code, notebooks (ingest to evaluation), audited splits and trained weights are released; three reproduction routes are documented (demo with pretrained weights, metric regeneration from released artifacts in ~ 10 minutes, and full retraining from public data). Datasets are downloaded from their official sources with checksum verification.

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